Teaching you IndicSent-M

1. What the project is

Sentiment analysis means deciding whether a piece of content is positive, negative, or neutral. **Multimodal** means using more than one type of input, here **text and images together**. **Indic** means Indian languages like Hindi, Bengali, Tamil, and Telugu, often mixed with English.

So the goal is a model that looks at a post (an image plus its caption or text) in an Indian language and predicts its sentiment.

2. Why text alone isn't enough

Take a meme showing a person crying, captioned "Wah, kya mausam hai 😂" under a flooded street. The text reads positive, but the image shows a bad situation, so the real sentiment is sarcastic or negative. A text-only model gets this wrong. A model that combines both signals can catch it.

Social media posts, memes, and product reviews with photos are largely multimodal, so this matters in practice.

3. Why Indian languages make it harder

- **Low-resource:** far less labeled data exists than for English.
- **Code-mixing:** people write "Hinglish" (Hindi in Roman script mixed with English), with no fixed spelling.
- **Many scripts:** Devanagari, Bengali, Tamil, and so on, each needing proper tokenization.
- **Cultural context:** sarcasm and humor depend on local references.

4. The pipeline, step by step

1. **Data:** collect image-text pairs, or use existing datasets, and label them with sentiment.
2. **Preprocessing:** for text, clean it, normalize scripts, and tokenize. For images, resize and normalize.
3. **Text encoder:** turns text into a vector of numbers capturing its meaning, using a multilingual model such as **MuRIL**, **IndicBERT**, or **XLM-R**.
4. **Image encoder:** turns the image into a vector, using **ViT**, **ResNet**, or **CLIP**'s image side.
5. **Fusion:** combines the two vectors (see below).
6. **Classifier:** a small neural layer that outputs the sentiment label.
7. **Evaluation:** compare against baselines.

5. Fusion, the core idea

- **Early fusion:** concatenate the features first, then classify. It is simple, but it can miss subtle interactions.

- **Late fusion:** classify text and image separately, then merge the predictions, for example by averaging. It is also simple, but the two modalities never interact.
- **Cross-attention fusion:** let the text "look at" parts of the image and vice versa. It is more powerful and can capture sarcasm, but it needs more data and compute.

A typical approach is to start with early and late fusion as baselines, then show that a better fusion method improves results.

6. How you prove it works

- **Baselines:** text-only model, image-only model, then your multimodal model.
- **Metrics:** accuracy and **macro-F1**. Macro-F1 averages the score across classes equally, so it is fairer when classes are imbalanced.
- **Hoped-for result:** multimodal beats both single-modality baselines, especially on cases where text and image disagree.
- **Error analysis:** show examples where the model fails, such as sarcasm or code-mixed slang.

7. Questions a panel is likely to ask

- *Why multimodal instead of text only?* Image and text can contradict or complement each other.
- *Why MuRIL or XLM-R and not plain BERT?* They are trained on Indian or multiple languages, so they handle those scripts and vocabulary.
- *How will you handle code-mixing?* Multilingual pretrained models plus text normalization.
- *Where is the data from?* Existing datasets, scraped public posts, or your own annotations. Be ready to explain labeling quality, for instance through multiple annotators.
- *Is it feasible?* Yes, with pretrained models, fine-tuning on Colab or Kaggle GPUs, and a modest dataset.
- *Why macro-F1?* It handles class imbalance.

8. Key terms to know

Embedding: a vector representing meaning. **Fine-tuning:** adapting a pretrained model to your task. **Transformer:** the architecture behind BERT and ViT. **Attention:** how a model weighs which parts of the input matter. **Transfer learning:** reusing knowledge from large pretraining.

I've assumed the standard approach here, since I don't know your team's exact plan on languages, datasets, or models. If you tell me those, I can tailor this, quiz you for practice, or build the PowerPoint. Which would help most?